THESIS TITLE

FULL NAME OF STUDENT

PhD

The Hong Kong Polytechnic University

20XX

Initial Submission for Examination Purpose

The Hong Kong Polytechnic University

Name of Department

Thesis Title

Full Name of Student

A thesis submitted in partial fulfilment of the requirements for the degree of
Doctor of Philosophy

Month 20XX

CERTIFICATE OF ORIGINALITY

I hereby declare that this thesis is my own work and that, to the best of my knowledge and belief, it reproduces no material previously published or written, nor material that has been accepted for the award of any other degree or diploma, except where due acknowledgement has been made in the text.

____________________	Full Name of Student
(Signed)	(Name of student)

Dedication

To my family.

Abstract

This thesis investigates a clearly defined research problem whose practical importance and unresolved scientific questions motivate the work. It begins by reviewing the relevant theoretical foundations and empirical evidence, identifying limitations in existing approaches and establishing the research questions addressed in the subsequent chapters. A reproducible methodology is then developed to examine these questions using appropriate data, experimental controls, evaluation measures, and statistical analyses. The proposed approach is compared with established baselines under consistent conditions, while additional experiments examine robustness, sensitivity to design choices, and the influence of the principal assumptions. The results demonstrate how the selected methodology addresses the stated research objectives and clarify the circumstances in which its advantages and limitations become most apparent. The discussion relates the findings to prior research, considers alternative explanations, and distinguishes conclusions supported directly by evidence from interpretations that require further validation. Particular attention is given to reproducibility, uncertainty, generalisability, ethical considerations, and the practical implications of the work. Taken together, the studies provide a coherent contribution to the field and establish a foundation for subsequent investigation. Future research should extend the evaluation to additional settings, test the approach using independent evidence, and examine how the identified limitations can be mitigated. Replace this anonymous example with a self-contained summary of the completed thesis, including its actual problem, methods, principal quantitative findings, conclusions, and significance.

Publications Arising from the Thesis

1. Author, A. A., and Collaborator, B. B. (20XX). Title of the first publication arising from the thesis. *Journal Name*, volume, pages.
2. Author, A. A., Collaborator, B. B., and Researcher, C. C. (20XX). Title of the second publication arising from the thesis. *Conference or Journal Name*, pages.

Acknowledgements

I gratefully acknowledge Name of Chief Supervisor, my Chief Supervisor, for guidance, advice, and supervision throughout this research. I also acknowledge the collaborators and colleagues whose contributions are identified in the relevant chapters and cited where appropriate. Replace this text before submission so that every contribution by supervisors, former supervisors, advisers, collaborators, technical staff, and others is declared accurately.

Contents

List of Figures

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List of Algorithms

List of Listings

List of Abbreviations

Abbreviation	Meaning
API	Application programming interface
DOI	Digital object identifier
PDF	Portable Document Format
RPg	Research postgraduate

CHAPTER 1

Introduction

1.1 Background

The introduction should establish the problem, its significance, and the context needed to understand the thesis. Begin with the broader research area, narrow the discussion to the specific problem, and define the scope of the investigation. Avoid turning this section into a second literature review; reserve detailed comparison of prior methods for Chapter 2.

Factual claims should be supported by appropriate sources. This sentence demonstrates a compressed numeric citation [1, 2]; citations appear as bracketed numbers such as [1] in the compiled thesis.

1.2 Aim and Objectives

State the overall aim in one precise sentence, then translate it into objectives that can be evaluated by the methods and results. Each objective should correspond to a later methodological step and a reported finding.

An objectives list can be used when it improves readability:

O1: define a reproducible problem setting;

O2: compare representative methods under consistent conditions; and

O3: analyse uncertainty, limitations, and generalisability.

1.2.1 Research Questions

Research questions should be specific, answerable, and aligned with the objectives. If hypotheses are used, state them after the relevant question and define the evidence that would support or challenge each one.

1.2.1.1 Scope and Terminology

Define the population, data, task, and evaluation boundary that apply throughout the thesis. Introduce specialised terms once and use them consistently; a glossary or abbreviation list is preferable when many recurring terms are required.

Technical terms may be clarified in a footnote.[1]

[1] Footnotes should be concise and should not be used as a substitute for essential argumentation or citation.

1.3 Thesis Organisation

Chapter 2 reviews the relevant literature. Chapter 3 demonstrates a self-contained study with its own introduction, methods, results, and discussion. Chapter 4 integrates the conclusions and identifies directions for future research.

CHAPTER 2

Literature Review

2.1 Conceptual Foundations

This section should introduce the theories, definitions, and assumptions on which the research depends. Organise the discussion by concepts or research questions rather than presenting one summary paragraph per paper. Where definitions vary, explain which interpretation the thesis adopts and why.

A strong review synthesises evidence across studies. It should identify areas of agreement, explain conflicting findings, and distinguish limitations in available data from limitations in methods or evaluation design.

2.2 Representative Approaches

Compare representative approaches using criteria that matter to the thesis, such as data requirements, assumptions, reproducibility, external validation, and computational cost. The comparison should make the rationale for the chosen methodology transparent rather than merely declaring one family of methods to be superior.

2.3 Research Gap

Conclude the review by identifying a focused and defensible gap. State what remains unknown, why existing evidence cannot resolve it, and how the proposed research addresses it. Do not define the gap only as the absence of a particular technique; connect it to a substantive research need.

As summarised in Table 2.1, the research gap should follow from a critical synthesis rather than a catalogue of publications.

Table 2.1. Example wide literature comparison with placeholder data. A short, plain optional caption is supplied for the List of Tables, while this longer caption may contain the detail needed to interpret the table.

Study	Setting	Data	Design	Primary metric	External test	Code	Key limitation
Example A [2]	Laboratory	Public	Retrospective	Accuracy	No	Yes	Single-site evaluation
Example B [1]	Simulation	Synthetic	Controlled	Error	Yes	No	Limited sensitivity analysis
Example C	Field study	Mixed	Prospective	Utility	Partial	Yes	Small sample size
Example D	Benchmark	Public	Cross-sectional	Score	No	Partial	Narrow task definition
Proposed study	Multiple settings	Mixed	Reproducible	Several metrics	Yes	Yes	Replace with actual limitations

CHAPTER 3

Study One: Example Empirical Study

3.1 Introduction

Open each study with the specific problem it addresses, the evidence motivating that problem, and the contribution made by this study. Keep the argument focused on the study rather than repeating the general introduction or literature review.

State the study objectives or hypotheses before presenting the methods. When a thesis contains additional studies, copy the complete `study-1` directory, rename its labels, and add the new study's `\input` line to `main.tex`.

3.2 Methods

3.2.1 Study Design

Describe the design in enough detail for a knowledgeable reader to reproduce the study. Record the sampling strategy, inclusion and exclusion criteria, variables, controls, preprocessing, model settings, software versions, random seeds, and evaluation protocol where applicable. Explain why the design can answer the research questions and identify any assumptions made before analysing the results.

The examples below demonstrate unified hyperlinks to Table 3.1, Figures 3.1 and 3.2, Equation (3.1), and Algorithm 3.1. Internal references, citations, and entries in the contents and lists use the same link treatment in the screen PDF.

Definition 3.1 (Example score)**.** For observations $s_1, \ldots, s_N$, the aggregate score is the arithmetic mean defined in Equation (3.1).

$$\text{Score} = \frac{1}{N}\sum_{i=1}^{N} s_i, \tag{3.1}$$

Proposition 3.2 (Bounded aggregate)**.** *If every observation satisfies $0 \le s_i \le 1$, then the score in Equation (3.1) also lies in $[0, 1]$.*

Proof. Summing the observation bounds gives $0 \le \sum_{i=1}^{N} s_i \le N$. Division by the positive integer N proves the stated result. □

For a vector prediction $\boldsymbol{y}$ and target $\boldsymbol{t}$, a composite objective may be typeset as

$$\mathcal{L}(\boldsymbol{y}, \boldsymbol{t}) = \lambda_1 \|\boldsymbol{y} - \boldsymbol{t}\|_1 + \lambda_2 \big(1 - \mathrm{sim}(\boldsymbol{y}, \boldsymbol{t})\big), \tag{3.2}$$

$$\lambda_1 + \lambda_2 = 1, \qquad \lambda_1, \lambda_2 \geq 0. \tag{3.3}$$

3.2.2 Data and Procedure

Summarise the origin and permitted use of each dataset, then document every split or transformation that could affect the conclusions. Use a table for compact dataset statistics and prose for decisions that require justification. A numbered algorithm is useful when the procedure has branching, iteration, or an ordering that must be reproduced.

Table 3.1. Example dataset summary with aligned numeric columns. All names and values are placeholders and must be replaced.

Dataset	Training	Validation	Test
Example Alpha	1000	200	300
Example Beta	2400	400	600
Example Gamma[a]	800	160	240

[a] This note demonstrates explanatory material that belongs directly with a table.

Algorithm 3.1. Example reproducible procedure

Require: data $\mathcal{D}$, maximum iterations T, tolerance ε
Ensure: fitted parameters $\boldsymbol{\theta}$

1: initialise $\boldsymbol{\theta}$ and record the random seed
2: **for** $t = 1, \ldots, T$ **do**
3: compute the objective and update $\boldsymbol{\theta}$
4: **if** the validation change is less than ε **then**
5: **break**
6: **end if**
7: **end for**
8: **return** $\boldsymbol{\theta}$

Listing 3.1. Example analysis code loaded from a separate source file.

```
from statistics import fmean

def mean_score(values):
    """Return the mean of a non-empty sequence."""
    if not values:
        raise ValueError("values must not be empty")
    return fmean(values)

def normalise(values):
    """Scale values by their largest absolute magnitude."""
    scale = max(abs(value) for value in values)
    return [value / scale for value in values]
```

3.2.3 Figures and Visual Evidence

Figures should remain legible at their final printed size and should not depend on colour alone. Supply descriptive captions, define symbols and abbreviations, and refer to every figure from the surrounding discussion. The examples below exercise a large full-width graphic and a multi-panel layout using standard `\includegraphics` commands. The example assets are stored under `figures/study-1/`; authors can replace either the files or their paths.

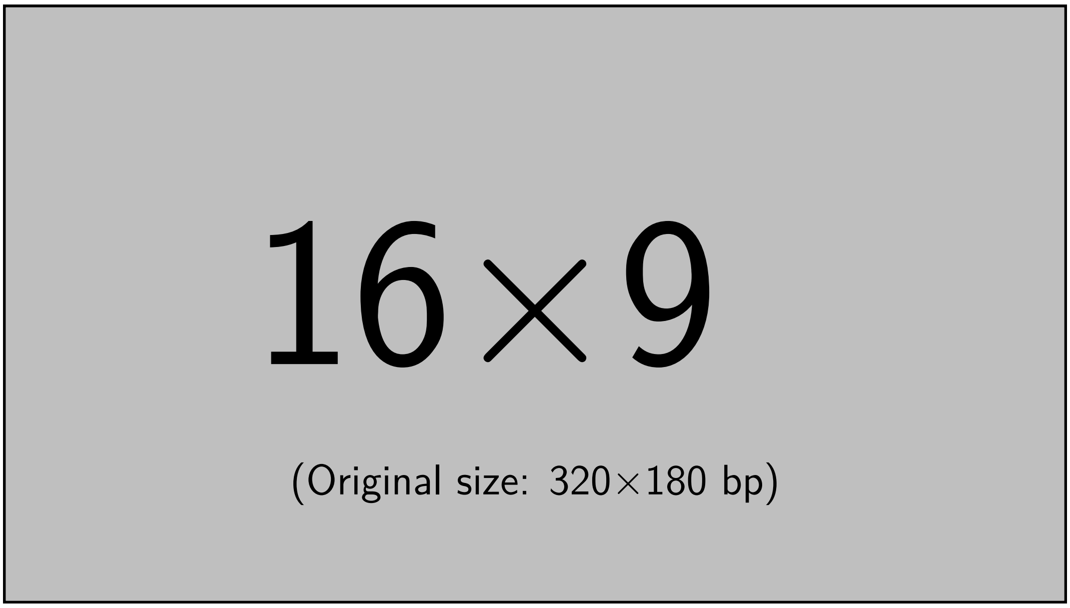
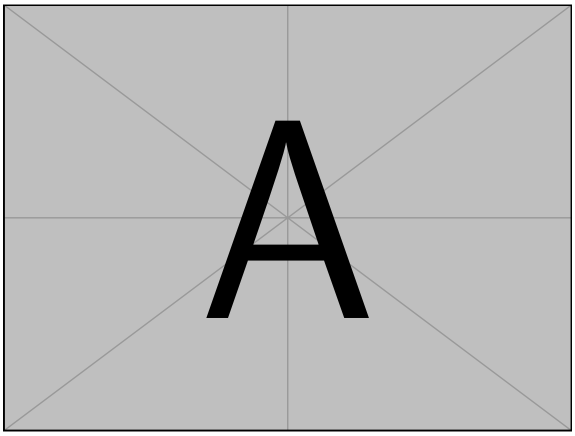
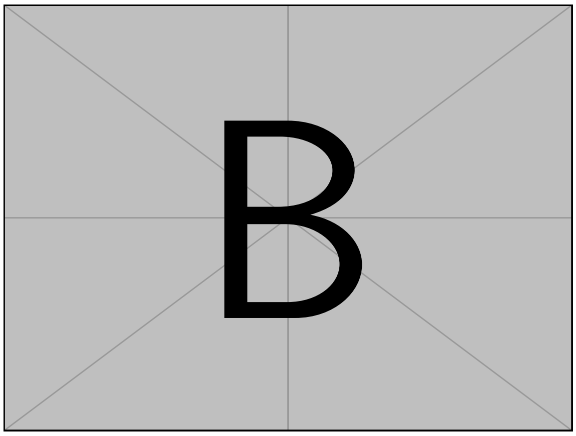


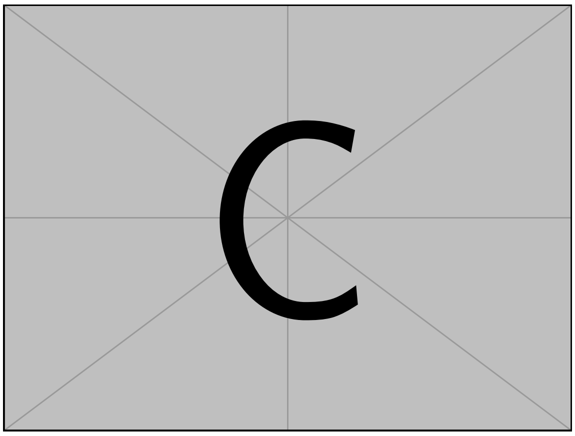
Figure 3.1. Example full-width figure included from a PDF asset. Replace the file path with a legible, high-resolution PDF, PNG, or JPEG image.



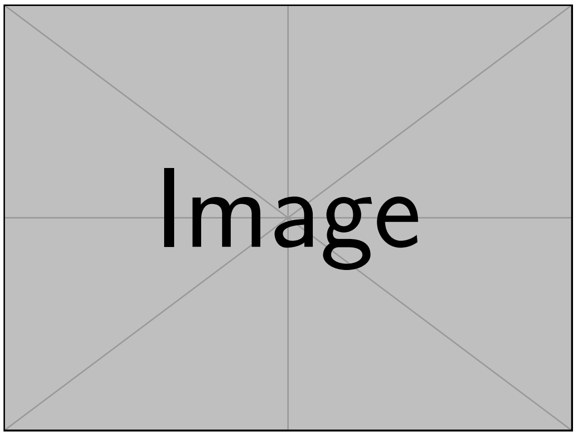
(a) Input



(b) Reference



(c) Baseline



(d) Proposed result

Figure 3.2. Example four-panel figure. This caption deliberately tests **bold best-result text** and underlined runner-up text; the plain optional caption keeps both styles out of the List of Figures.

3.3 Results

Present results in the same order as the research questions or objectives. Report sample sizes, uncertainty, and the statistical procedure alongside point estimates. Use tables for exact comparisons and figures for patterns; the prose should explain the important finding rather than repeat every displayed value.

Table 3.2. Example ranked result table with fictitious values. The caption tests **bold best-result text** and <u>underlined runner-up text</u>; the plain optional caption keeps both styles out of the List of Tables. Arrows indicate the preferred direction of each metric.

Method	**Primary evaluation**		**External evaluation**	
	Score ↑	**Error ↓**	**Score ↑**	**Time (ms) ↓**
Baseline A	0.812 ± 0.011	0.184 ± 0.008	0.774	18.2
Baseline B	0.846 ± 0.009	<u>0.151 ± 0.006</u>	0.801	**12.7**
Baseline C	<u>0.873 ± 0.007</u>	0.158 ± 0.007	<u>0.829</u>	16.4
Proposed method	**0.891** ± **0.006**	**0.143** ± **0.005**	**0.851**	<u>14.1</u>

Values are illustrative means ± standard deviations. Replace them with verified results and state the number of repetitions and statistical procedure.

Use `\best{...}` and `\runnerup{...}` throughout the thesis rather than encoding rank with colour alone. This remains legible when printed in greyscale and keeps the meaning consistent across tables.

3.4 Discussion

Interpret the results in relation to the study objectives and the literature. Use ablation studies to isolate important design choices and robustness analyses to test sensitivity to plausible alternatives. Report negative or inconclusive findings when they affect the interpretation.

The interpretation should connect Table 3.2 to the research questions, report uncertainty, and distinguish statistical evidence from practical significance. Discuss study-specific limitations and keep claims within the population, data, and conditions actually evaluated.

CHAPTER 4

Conclusions and Suggestions for Future Research

4.1 Conclusions

The conclusion should answer the thesis objectives directly, identify the contribution supported by each study, and explain the overall significance of the findings. Keep the claims proportional to the evidence and avoid introducing new experiments, citations, or arguments at this stage.

4.2 Limitations

Summarise the constraints that materially affect validity or generalisability. For each important limitation, explain whether it may change the magnitude, direction, or applicability of the conclusions and note any mitigation already incorporated into the study design.

4.3 Suggestions for Future Research

Future work should follow from unresolved questions in the results and limitations. Prioritise concrete studies that could validate, extend, or challenge the thesis findings, and distinguish near-term follow-up experiments from broader long-term research directions.

APPENDIX A

Supplementary Material

Appendices are optional. This example demonstrates a table that repeats its heading across several pages. Remove this file and its `\input` line from `main.tex` when no appendix is required.

A.1 Multi-page Record

Table A.1. Example multi-page supplementary table containing distinct reproducibility records and a repeated header.

ID	Record	Information to report
01	Data provenance	Identify the source, collection period, licence, version, and any access restrictions for every dataset used in the thesis.
02	Cohort definition	Record inclusion and exclusion criteria, recruitment flow, and the final number of observations available for each analysis.
03	Split protocol	State how training, validation, and test partitions were formed and how leakage between related observations was prevented.
04	Preprocessing	List transformations in execution order, including normalisation constants, resizing rules, filtering, and handling of invalid records.
05	Hyperparameters	Report the search space, selection criterion, final settings, and whether tuning used information from the held-out test set.
06	Randomness	Provide random seeds and identify operations whose output may remain non-deterministic across hardware or software versions.
07	Ablation study	Define the component removed or altered, the controlled baseline, and the metric used to attribute a change in performance.

Continued on the next page

Table A.1 continued from the previous page

ID	Record	Information to report
08	External validation	Describe domain differences between development and external data and state whether any adaptation occurred before evaluation.
09	Missing data	Give the amount and pattern of missingness, the chosen treatment, and a sensitivity analysis when assumptions may affect conclusions.
10	Calibration	Specify the calibration measure, binning or fitting procedure, confidence intervals, and any post-hoc adjustment applied to predictions.
11	Resource use	Report timing boundaries, repetitions, hardware, batch size, numerical precision, and peak host and accelerator memory.
12	Software environment	Preserve operating-system, compiler, library, and driver versions together with a machine-readable environment specification.
13	Ethics and consent	Record approving bodies, reference numbers, consent or waiver arrangements, and safeguards for privacy-sensitive material.
14	Artifact archive	Provide persistent locations, checksums, licences, and access instructions for code, configurations, trained models, and derived data.

A.2 Supplementary Derivation

Use an appendix for supporting material that is necessary for verification but would interrupt the main argument, such as extended derivations, additional experiments, questionnaires, or implementation details. Give every appendix item a clear purpose and refer to it from the main text when it supports a claim.

$$\widehat{\theta} = \arg\min_{\theta \in \Theta} \sum_{i=1}^{N} \ell\big(f_\theta(x_i), y_i\big). \tag{A.1}$$

References

[1] F. Author and S. Collaborator. “Example Article Title”. In: *Journal Name* 1.1 (2025), pp. 1–10.

[2] A. Researcher and B. Scholar. “A Fictitious Conference Paper for Demonstrating Numeric Citations”. In: *Proceedings of the Example Conference*. 2024, pp. 11–20.